

Mars Reconnaissance Orbiter Navigation Strategy for Support of Mars 2020 Mission's Entry, Descent and Landing Sequence

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Mars Reconnaissance Orbiter Project



THE UNIVERSITY OF ARIZONA,



Mars Reconnaissance Orbiter (Mission, Spacecraft and PSO)

MRO launched in August 2005 from Cape Canaveral Air Force Station arrived at Mars in March 2006 & started science operations in November 2006. It has nearly completed 16 years since launch (70,000 orbits by July 2, 2021) and to date has returned over 416 Terabytes of data.

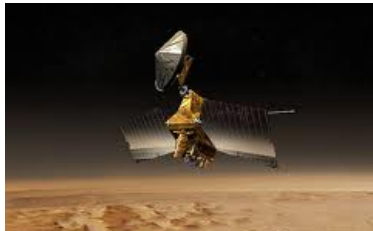


MRO Primary Science Orbit (PSO):

- **Sun-synchronous** orbit ascending node at 3:00 PM $\pm$ 15 minutes Local Mean Solar Time (LMST) (daylight equatorial crossing)
- Periapsis is **frozen** about the Mars South Pole
- **Near-repeat ground track walk (GTW)** every $\sim$ 17-day, 211 orbit (short-term repeat) MRO targeting cycle, exact repeat after 4602 orbits. The nominal GTW is 32.45811 km West each 211 orbit cycle (maintained within $\pm$ 60 km with periodic maneuvers)

MRO Spacecraft:

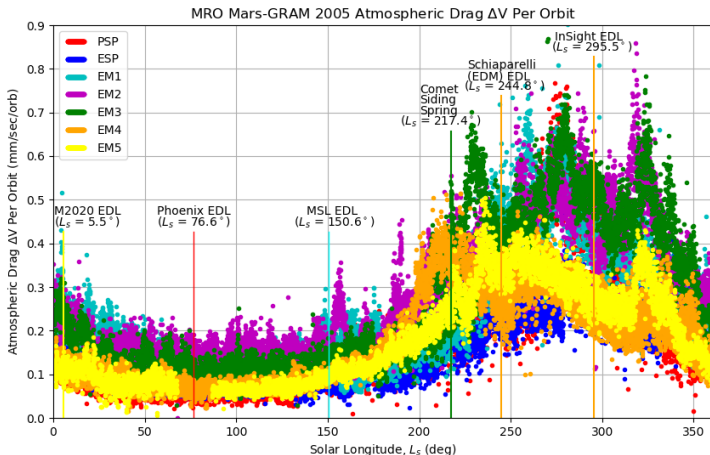
- **Spacecraft Bus:** 3-axis stabilized ACS system; 3-meter diameter High Gain Antenna; hydrazine propulsion system
- **Instrument Suite:** HiRISE Camera, CRISM Imaging spectrometer, Mars Climate Sounder, Mars Color Imager, Context Camera, Shallow Subsurface Radar, Electra engineering payload (among other instrument payloads)



Navigation Plan

- MRO Navigation Team was tasked to develop the strategy to phase MRO to support Mars 2020 (M2020) EDL
 - EDL relay support
 - EDL relay support using UHF antenna (pseudo bent-pipe)
 - Maintain orbit for 7 sols of uninterrupted overflight support
 - Subsequent surface operations support
 - Phasing Target
 - Per target given in EDL Relay Target File (ERTF)
 - Time to cross a set latitude
 - Optimized to support EDL
 - Navigation Requirement: ± 30 seconds
 - Navigation Plan
 - Phasing offset just prior to first phasing maneuver was about 65 minutes (early)
 - Above 1/2 of the orbit period (~ 112 minutes)
 - 4 phasing maneuvers were planned, but 2 implemented
 - Phasing correction in accordance with Navigation uncertainty

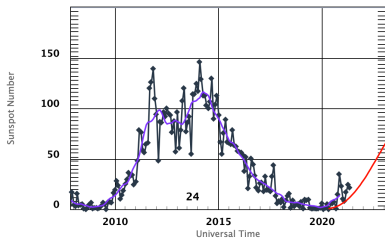
Mars Atmospheric Density Variation



- **Atmospheric density variation is a significant navigation error source**
- **Anticipated drag ΔV :** 0.20-0.30 mm/s/orbit at the time of maneuvers leading to M2020 landing
- **Slightly larger than earlier phasing support:** MSL EDL (08/2012) & Phoenix EDL (03/2008)

Solar Cycle and Density Scale Factor

ISES Solar Cycle Sunspot Number Progression



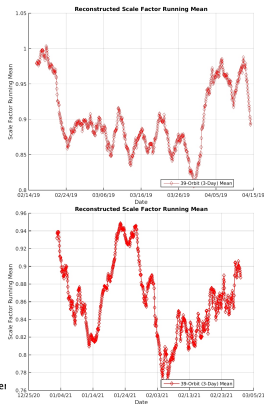
◆ Monthly Values — Smoothed Monthly Values — Predicted Values
Space Weather Prediction Center

• Solar Cycle:

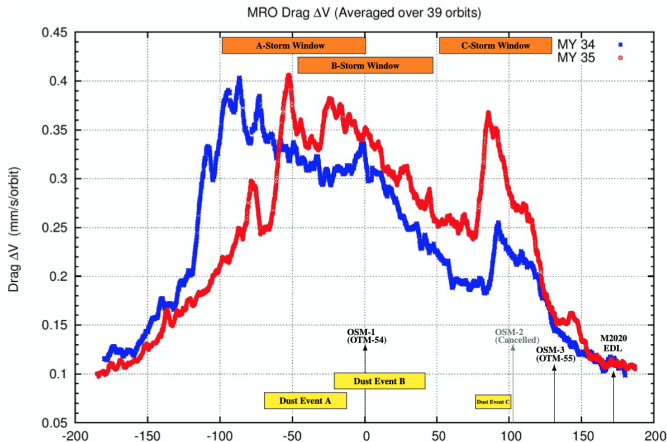
- Near solar minimum
- Low volatility in the atmospheric density variations

• Density Scale Factor (DSF):

- Considered DSF behavior from previous Mars year (MY 34)
- Average DSF 0.90 was selected for the final maneuver



Atmospheric Drag ΔV



- **Maneuver ΔV :** Pro-velocity decreases effective drag on the S/C
- **Dust Activity:** Dust storm adds to the drag ΔV
- Overall drag effects comparable to previous Mars year

Maneuvers and Phasing Corrections

	OSM-1 (OTM-54)	OSM-2	OSM-3 (OTM-55)	M2020 EDL Target
OD DCO for OSM	8/23/2020	12/07/2020	1/11/2021	
DCO Prior to M2020 EDL Target	179.3 days	73.2 days	38.4 days	
Average Drag ΔV Per Orbit	0.30 mm/s	0.25 mm/s	0.20 mm/s	
OD DCO Down-Track Timing Uncertainty (3σ)	24.6 min	3.5 min	47.4 sec	
Post-OSM Down-Track Timing Uncertainty (3σ)	21.8 min		27.1 sec	
OSM/EDL Date	9/02/2020	12/16/2020	1/20/2021	2/18/2021
OSM Execution Prior to M2020 EDL	169.3 days	64.3 days	29.2 days	
OSM Design ΔV	0.198930 m/s	<i>cancelled</i>	0.269661 m/s	
OSM Reconstructed ΔV	0.203020 m/s		0.270138 m/s	
M2020 EDL Relay Target File	ERTF-6		ERTF-10	ERTF-10
Predicted Phasing Correction*	44.3 min early		9.83 min late	± 30 sec
Predicted Phasing Offset Remaining†	24.0 min early		0.0 min	requirement
Actual Phasing Offset (Pre-OSM)	64.5 min early		9.86 min late	
Actual Phasing Correction†	45.6 min early		9.86 min late	Achieved:
Actual Phasing Offset (Post-OSM)	18.9 min early		0.52 sec late	1.97 sec late
3:30 PM LMST Target @ EDL (3:15–3:35 PM Req.)	3:30:27 PM		3:30:36 PM	3:30:36 PM

* Based on maneuver design (does not account for OD drift from DCO to OSM implementation or maneuver execution errors)

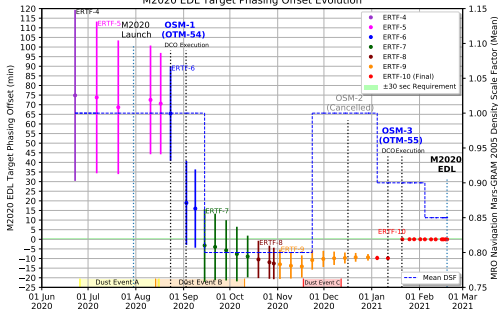
† Accounts for down-track timing uncertainty (3σ)

‡ Based on maneuver execution (accounts for OD drift from DCO to OSM implementation and maneuver execution errors)

- **Phasing Maneuvers:** OSM-1 (9/02/2020) and OSM-3 (1/20/2021)
- **Post-OSM-3 phasing offset:**
 - Grew to -1.97 seconds (ERTF-10) at the time of Perseverance landing
 - Actual offset was 6.46 seconds (ERTF-15)

Phasing Offset History

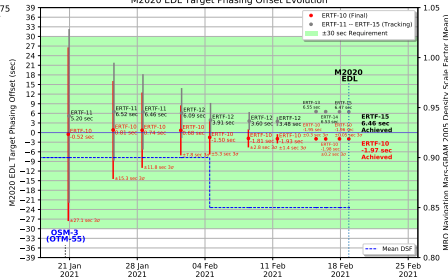
M2020 EDL Target Phasing Offset Evolution



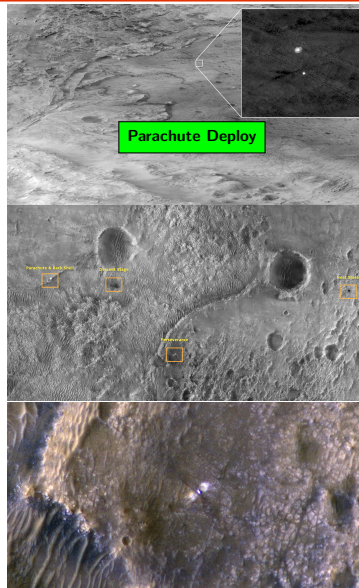
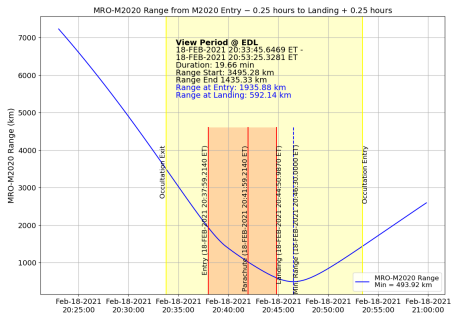
- **Phasing offset monitoring:** Started from 241 days prior to M2020 landing (38 days prior to M2020 launch)
- **Navigation uncertainty at OD DCO:** 24.6 min (OSM-1), 47.4 sec (OSM-3)
- **Final phasing offset:** 1.97 sec $\ll$ 30 seconds requirement

- **First Phasing Maneuver:** OSM-1 (OTM-54) on September 2, 2020
 - Designed to remove about 44 minutes of the predicted phasing offset
- **Second Phasing Maneuver:** OSM-3 (OTM-55) on January 20, 2021
 - Designed to remove remaining offset of about 9.83 minutes

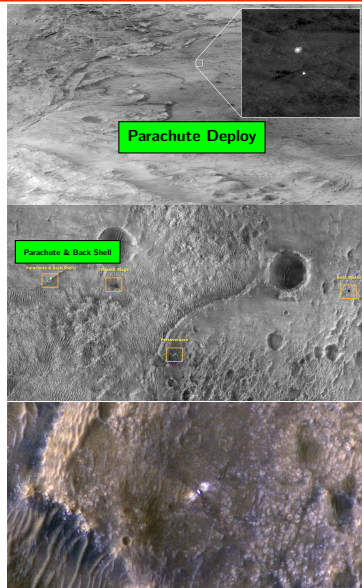
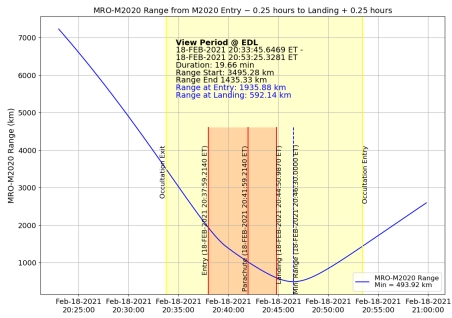
M2020 EDL Target Phasing Offset Evolution



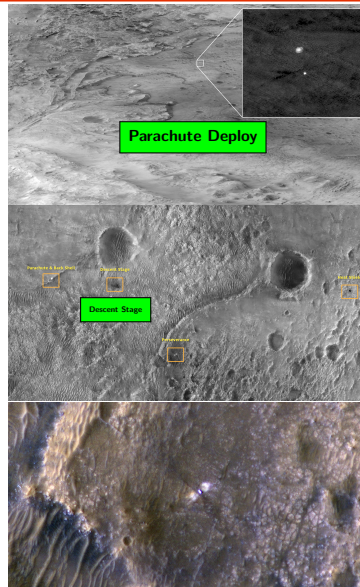
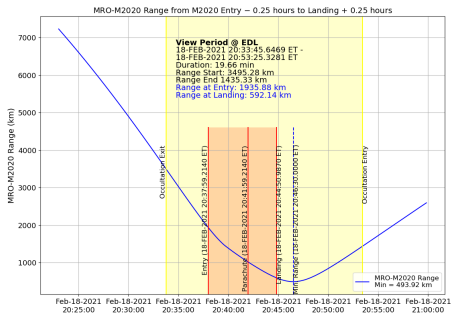
M2020 Landing (Images Taken by HiRISE)



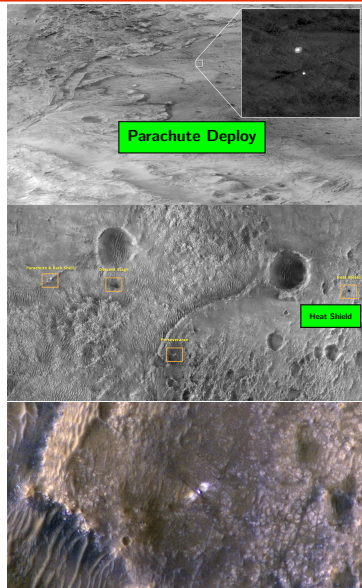
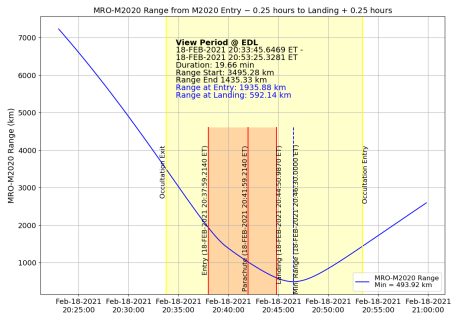
M2020 Landing (Images Taken by HiRISE)



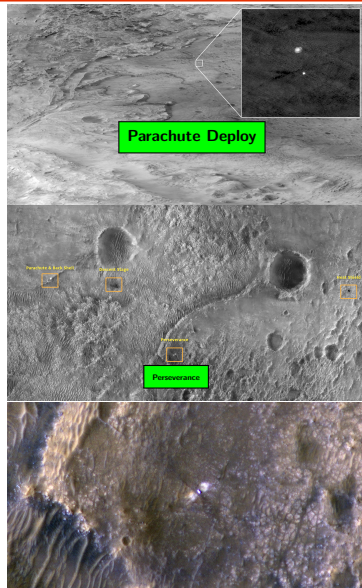
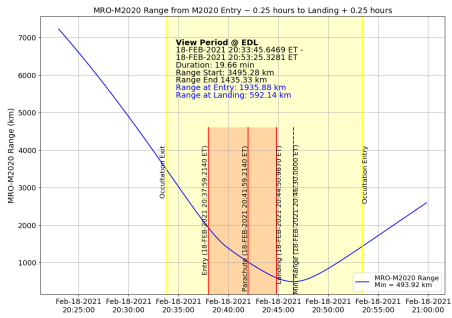
M2020 Landing (Images Taken by HiRISE)



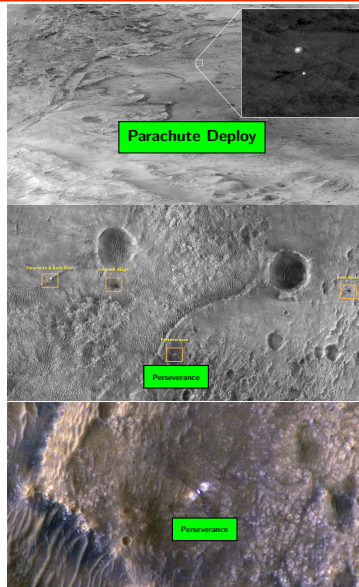
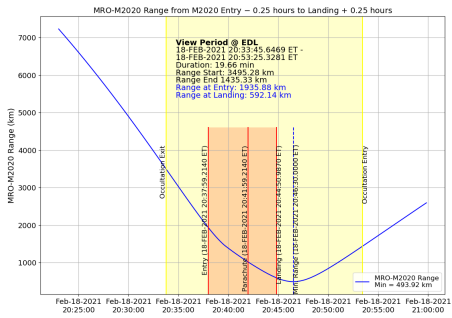
M2020 Landing (Images Taken by HiRISE)



M2020 Landing (Images Taken by HiRISE)



M2020 Landing (Images Taken by HiRISE)



Summary

- MRO was successfully phased to support M2020 EDL
 - Final phasing offset was 1.97 seconds late (ERTF-10)
 - Actual phasing offset was 6.46 seconds early (ERTF-15)
 - Well within the ± 30 seconds timing requirement given in ERTF-10 (final target)
 - Accurate phasing led to good EDL relay support by Electra
 - Via bent-pipe telecommunications
 - Parachute deploy imaged
- Post-landing observations done by HiRISE camera
 - Perseverance rover, parachute, back shell, & heat shield
- MRO returning to nominal Primary Science Orbit (PSO)
 - OTM-57 performed on June 9, 2021 to return to 3:10 PM LMST
 - Resumed MRO science operations and surface operations support to landers

Backup Slides

M2020 ERTF-10 and ERTF-15

EDL RELAY TARGETS FILE (ERTF)

Data generated on Mon Jan 4 14:53:39 2021 by J. Kangas,
NASA Jet Propulsion Laboratory

M2020 SPK File:

m2020_trajCEDLS-6DOF_od068v1_l4_20210218-20210219.bsp

MRO SPK File: m2020_spk_mro_ertf-10_od068v1_74.bsp

```
*****
* MRO RELAY TARGETS
* Epoch                  : 18-FEB-2021 20:37:56.8164 ET
* LMST (Asc.Eq.Cross)    : 15:30:40
* Latitude (2000 IAU Mars Fixed) : -9.6220 deg
*****
```

M2020 Data (2000 IAU Mars Fixed)

```
Entry Epoch      : 18-FEB-2021 20:37:56.8164 ET
Entry Latitude   : 20.7753 deg
Entry Longitude  : 66.7613 deg
Entry Radius     : 3522.2000 km
Entry Range to MRO : 1934.6677 km
Landing Epoch   : 18-FEB-2021 20:44:48.6730 ET
Landing Latitude : 18.4552 deg
Landing Longitude : 77.4112 deg
Landing Radius   : 3391.9670 km
Landing Range to MRO : 592.5212 km
```

MRO Data - Keplerian Osculating Elements (2000 IAU Mars Pole)

```
Entry Epoch      : 18-FEB-2021 20:37:56.8164 ET
Semimajor Axis   : 3663.7693 km
Eccentricity     : 0.009255
Inclination      : 92.8182 deg
RAAN             : 271.5197 deg
Arg. of Periapsis : 275.3434 deg
Mean Anomaly     : 74.0001 deg
True Anomaly     : 75.0228 deg
```

MRO Data - Spherical Coordinates (2000 IAU Mars Fixed)

```
Entry Epoch      : 18-FEB-2021 20:37:56.8164 ET
Target Latitude  : -9.6220 deg
Target Longitude : 73.9461 deg
LMST (Asc.Eq.Cross) : 15:30:40
LTST (Asc.Eq.Cross) : 14:52:44
Landing Epoch   : 18-FEB-2021 20:44:48.6730 ET
MRO Lat. at Landing : 12.4029 deg
MRO Lon. at Landing : 71.1750 deg
```

(a) M2020 ERTF-10 (Used for Final MRO Phasing Maneuver)

EDL RELAY TARGETS FILE (ERTF)

Data generated on Wed Feb 17 07:53:26 2021 by J. Kangas,
NASA Jet Propulsion Laboratory

M2020 SPK File:

m2020_trajCEDLS-6DOF_od109v1_NOBURN_20210218-20210219.bsp

MRO SPK File: m2020_spk_mro_ertf-15_od109v1_74.bsp

```
*****
* MRO RELAY TARGETS
* Epoch                  : 18-FEB-2021 20:37:59.2140 ET
* LMST (Asc.Eq.Cross)    : 15:30:40
* Latitude (2000 IAU Mars Fixed) : -9.9457 deg
*****
```

M2020 Data (2000 IAU Mars Fixed)

```
Entry Epoch      : 18-FEB-2021 20:37:59.2140 ET
Entry Latitude   : 20.8274 deg
Entry Longitude  : 66.7975 deg
Entry Radius     : 3522.2000 km
Entry Range to MRO : 1956.4132 km
Landing Epoch   : 18-FEB-2021 20:44:50.9870 ET
Landing Latitude : 18.4548 deg
Landing Longitude : 77.4213 deg
Landing Radius   : 3391.9579 km
Landing Range to MRO : 605.5539 km
```

MRO Data - Keplerian Osculating Elements (2000 IAU Mars Pole)

```
Entry Epoch      : 18-FEB-2021 20:37:59.2140 ET
Semimajor Axis   : 3663.7893 km
Eccentricity     : 0.008950
Inclination      : 92.8183 deg
RAAN             : 271.5210 deg
Arg. of Periapsis : 275.0534 deg
Mean Anomaly     : 73.9999 deg
True Anomaly     : 74.9888 deg
```

MRO Data - Spherical Coordinates (2000 IAU Mars Fixed)

```
Entry Epoch      : 18-FEB-2021 20:37:59.2140 ET
Target Latitude  : -9.9457 deg
Target Longitude : 73.9541 deg
LMST (Asc.Eq.Cross) : 15:30:40
LTST (Asc.Eq.Cross) : 14:52:44
Landing Epoch   : 18-FEB-2021 20:44:50.9870 ET
MRO Lat. at Landing : 12.0737 deg
MRO Lon. at Landing : 71.1839 deg
```

(b) M2020 ERTF-15 (Final Verification)

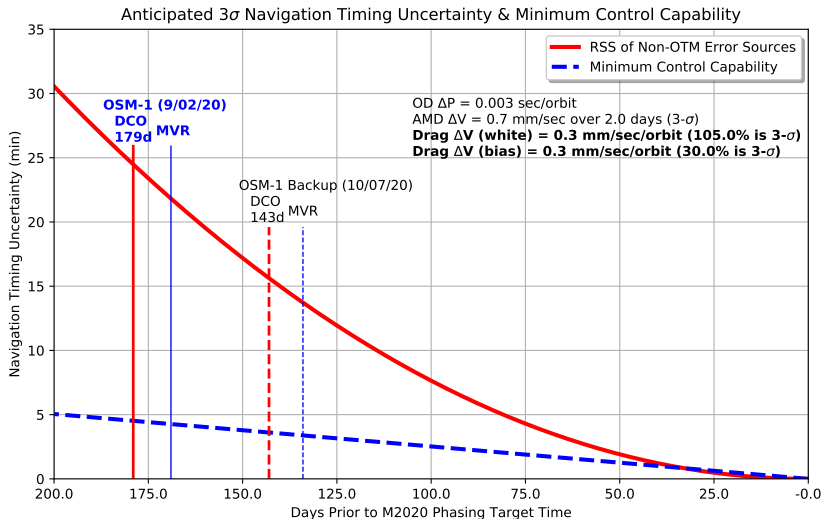
Maneuver ΔV and Direction

Maneuver	Maneuver Epoch (UTC SCET)	Orbital Apsis/ Node	Data Source	ΔV (mm/s)		Right Ascension (deg)		Declination (deg)		Duration (sec)	
				Value	err	Value	err	Value	err	Dur	err
OTM-52 (OCM-4)	12-Dec-2018 14:18:23	DEqX	Recon	1412.1	9.0	2.66	1.43	-60.47	0.03	20.1	0.2
			Design	1403.0		1.23		-60.50		20.3	
OTM-53 (OCM-4A)	23-Oct-2019 12:15:28	AEqX	Recon	2803.1	10.0	329.86	0.93	-30.76	0.38	39.3	0.4
			Design	2793.1		328.93		-30.39		39.7	
OTM-54 (OSM-1)	02-Sep-2020 13:41:38	Apo	Recon	203.0	4.1	47.71	0.16	-0.52	0.30	9.4	0.4
			Design	198.9		47.87		-0.82		9.0	
OSM-2	16-Dec-2020	 Cancelled									
OTM-55 (OSM-3)	20-Jan-2021 14:36:48	Apo	Recon	270.1	0.5	300.03	0.15	-35.70	0.17	11.8	0.4
			Design	269.7		299.88		-35.53		12.2	
OSM-4	04-Feb-2021	 Contingency maneuver if ± 30 sec requirement not met									
OSM-4C	10-Feb-2021	 Contingency maneuver if ± 30 sec requirement not met									
M2020 EDL Target Time: 18-Feb-2021 20:37:56.8164 ET SCET											
OTM-56	14-Apr-2021 11:44:29	Peri	Recon	814.0	10.7	356.10	0.10	-33.85	0.40	11.7	0.1
			Design	803.3		356.20		-33.45		11.8	
OTM-57 (OCM-5)	09-Jun-2021 13:19:53	DEqX	Recon	19504.3	0.7	282.02	0.14	-35.59	0.02	271.0	4.4
			Design	19503.6		281.88		-35.61		275.4	

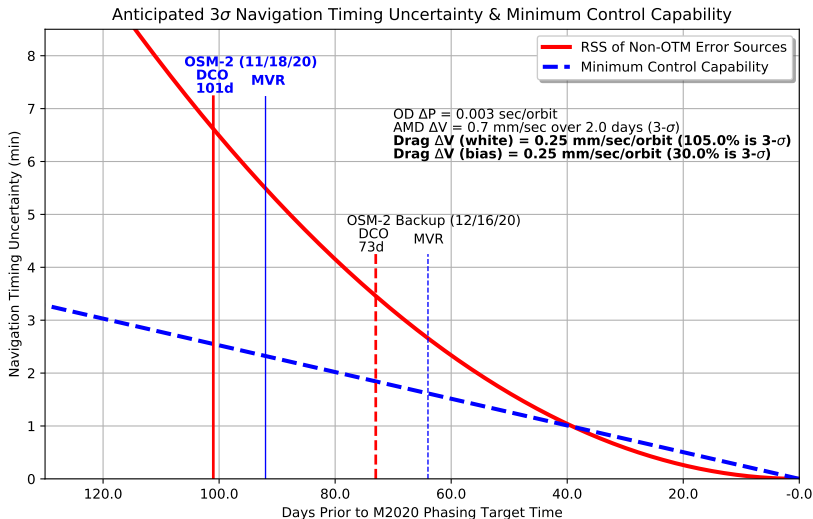
Evolution of MRO Phasing Offset from M2020 EDL Target

OD Data Cutoff (ET)	Entry Relay Target File (ERTF)	Days to EDL from DCO	Avg. DSF	Avg. Drag ΔV Per Orbit	Navigation Timing Uncertainty (3σ)	EDL Phasing Offset	
						Convention: + (early), - (late) Targeted	Tracking (ERTF)
22-JUN-2020 14:15:00	ERTF-4	241.3	1.00	0.30	44.47 min	+74.85 min	
06-JUL-2020 14:45:00	ERTF-5	227.2	1.00	0.30	39.45 min	+73.79 min	
20-JUL-2020 14:40:00	ERTF-5	213.2	1.00	0.30	34.77 min	+68.67 min	
10-AUG-2020 12:20:00	ERTF-5	192.3	1.00	0.30	28.27 min	+72.51 min	
17-AUG-2020 02:45:00	ERTF-5	185.7	1.00	0.30	26.37 min	+70.58 min	
23-AUG-2020 12:33:00	ERTF-6	179.3	1.00	0.30	24.57 min	+65.40 min	
OSM-1/OTM-54 (02-SEP-2020 13:42:47.5613 ET): designed to remove 44.3 min of offset (24.0 min remaining)							
02-SEP-2020 19:00:00	ERTF-6	169.1	1.00	0.30	21.84 min	+18.93 min	
08-SEP-2020 13:37:00	ERTF-6	163.3	1.00	0.30	20.37 min	+15.98 min	
14-SEP-2020 14:40:00	ERTF-7	157.2	0.80	0.30	18.89 min	-3.21 min	
21-SEP-2020 10:10:00	ERTF-7	150.4	0.80	0.30	17.29 min	-3.97 min	
28-SEP-2020 14:09:30	ERTF-7	143.3	0.80	0.30	15.68 min	-5.74 min	
05-OCT-2020 12:35:00	ERTF-7	136.3	0.80	0.30	14.20 min	-7.67 min	
12-OCT-2020 12:47:00	ERTF-7	129.3	0.80	0.25	10.85 min	-8.94 min	
19-OCT-2020 12:20:00	ERTF-8	122.3	0.80	0.25	9.71 min	-10.47 min	
26-OCT-2020 15:10:00	ERTF-8	115.2	0.80	0.25	8.61 min	-11.93 min	
29-OCT-2020 11:35:00	ERTF-8	112.4	0.80	0.25	8.19 min	-12.57 min	
02-NOV-2020 11:15:00	ERTF-9	108.4	0.80	0.25	7.62 min	-13.01 min	
09-NOV-2020 13:46:00	ERTF-9	101.3	0.80	0.25	6.65 min	-13.75 min	
16-NOV-2020 15:44:00	ERTF-9	94.2	0.80	0.25	5.76 min	-14.12 min	
23-NOV-2020 09:55:00	ERTF-9	87.4	1.00	0.25	4.96 min	-10.82 min	
30-NOV-2020 14:30:00	ERTF-9	80.3	1.00	0.25	4.18 min	-10.20 min	
07-DEC-2020 14:39:00	ERTF-9	73.2	1.00	0.25	3.48 min	-9.87 min	
14-DEC-2020 14:15:00	ERTF-9	66.3	1.00	0.25	2.85 min	-9.72 min	
21-DEC-2020 14:00:00	ERTF-9	59.3	1.00	0.25	2.28 min	-9.38 min	
29-DEC-2020 14:00:00	ERTF-9	51.3	1.00	0.25	1.71 min	-9.37 min	
04-JAN-2021 13:23:00	ERTF-10	45.3	0.90	0.20	1.10 min	-9.71 min	
11-JAN-2021 12:05:00	ERTF-10	38.4	0.90	0.20	47.4 sec	-9.84 min	
OSM-3/OTM-55 (20-JAN-2021 14:37:57.8233 ET): designed to remove remaining offset							
20-JAN-2021 20:52:00	ERTF-10	29.0	0.90	0.20	27.1 sec	-0.52 sec	ERTF-11: +5.20 sec
25-JAN-2021 12:10:00	ERTF-10	24.0	0.90	0.15	15.3 sec	+0.81 sec	ERTF-11: +6.52 sec
28-JAN-2021 12:06:30	ERTF-10	21.4	0.90	0.15	11.8 sec	+0.74 sec	ERTF-11: +6.46 sec
01-FEB-2021 12:00:00	ERTF-10	17.4	0.90	0.15	7.8 sec	+0.68 sec	ERTF-12: +6.09 sec
04-FEB-2021 11:41:00	ERTF-10	14.4	0.85	0.15	5.3 sec	-1.50 sec	ERTF-12: +3.91 sec
08-FEB-2021 11:38:00	ERTF-10	10.4	0.85	0.15	2.8 sec	-1.81 sec	ERTF-12: +3.60 sec
11-FEB-2021 11:34:30	ERTF-10	7.4	0.85	0.15	1.4 sec	-1.93 sec	ERTF-12: +3.48 sec
15-FEB-2021 11:25:00	ERTF-10	3.4	0.85	0.15	0.3 sec	-1.95 sec	ERTF-13: +6.55 sec
16-FEB-2021 11:20:00	ERTF-10	2.4	0.85	0.15	0.2 sec	-1.98 sec	ERTF-14: +6.53 sec
17-FEB-2021 21:30:00	ERTF-10	1.0	0.85	0.15	0.05 sec	-1.96 sec	ERTF-15: +6.47 sec
M2020 EDL Phasing Target Reconstruction			0.85	0.15	per ERTF-10:	-1.97 sec	ERTF-15: +6.46 sec

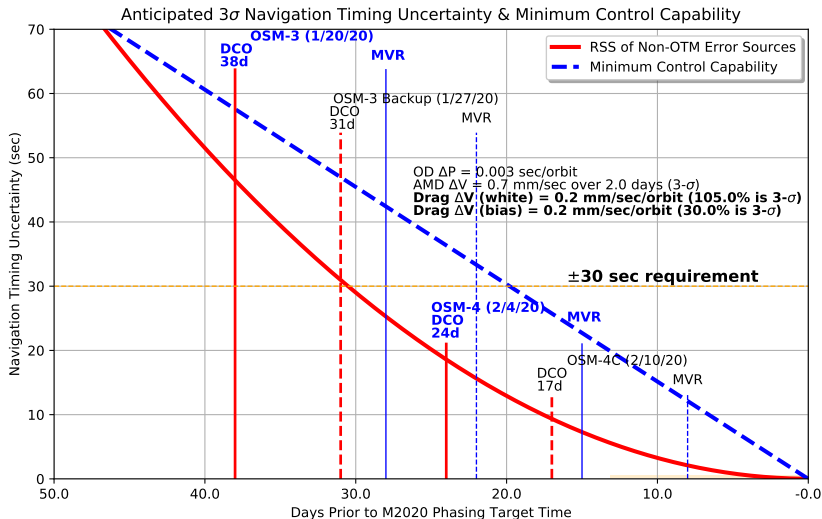
Nav. Timing Unc. ($3\text{-}\sigma$) for OSM-1 (Drag $\Delta V = 0.30$ mm/s/orbit)



Nav. Timing Unc. ($3\text{-}\sigma$) for OSM-2 (Drag $\Delta V = 0.25$ mm/s/orbit)



Nav. Timing Unc. (3- σ) for OSMs 3 & 4 (Drag $\Delta V = 0.20$ mm/s/orbit)



Ground Track Walk Error at M2020 EDL



- OTM-54 (pro-velocity) & OTM-55 (anti-velocity) to achieve M2020 EDL target
 - GTW error grew to about +355 km by OTM-56
- Inclination-change maneuver (OTM-57) to achieve 3:10 PM LMST for PSO by November 2021



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